BUTTERFLYFX ANPC DESIGN SUITE

Starfighter Launch Bay & Mission Cycle Design Document

BFXG-009 — Document #9 in the Suite

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Category	Instructions / Report Hybrid
Audience	Game Developers & AI Coding Assistants

> *"Store Geometry Not Bytes"*
>
> — ButterflyFX Design Philosophy

PURPOSE

This document defines the **complete round-by-round game loop** — every phase the player experiences from mission briefing through combat, return, reward, and relaunch. This cycle repeats every round and constitutes the core gameplay experience of the Starfighter game. Every system described herein must be implemented for the mission cycle to function.

Concept Reference — Launch Bay Interior aboard the HumanSpaceBattleShip carrier

1. Overview — The Mission Cycle

The Starfighter game loop is built on a **10-phase repeating mission cycle**. Every round follows this exact sequence. The cycle begins when the player is seated in the launch tube of the HumanSpaceBattleShip carrier and ends when the player is back in the launch tube, refitted and briefed for the next sortie. The loop continues until a Game Over condition is triggered or the player completes the final round.

1.1 Mission Cycle Flow Diagram

Phase	Name	Summary	Player Control
1	**Launch Bay Briefing**	Announcer delivers threat briefing, objectives, loadout check, clearance to launch	Listen only
2	**Launch Sequence**	Cut scene — launch tube doors open, fighter catapults out of carrier, HUD activates	Watch (skippable)
3	**Active Combat / Mission**	Player engages enemies,	**Full control**

Phase	Name	Summary	Player Control
		completes primary and secondary objectives	
4	**In-Combat Emergency Systems**	Rescue Shuttle, Fuel Tanker, Countermeasures — triggered as needed during Phase 3	**Full control**
5	**Mission Complete**	Announcer confirms objectives met, autopilot engages, ship returns to carrier	Watch only
6	**Docking Cut Scene**	Ship approaches carrier, enters docking bay, lands, is moved to launch tube	Watch (skippable)
7	**Refit & Reward**	Debrief, performance summary, XP/credits, ability/weapon unlocks, rank promotion	Review & interact
8	**Test Flight**	Optional safe environment to practice new abilities/weapons	**Full control** (optional)
9	**Next Mission Briefing**	Announcer delivers next round's briefing — new threats, objectives, loadout	Listen only
10	**Launch — Cycle Repeats**	Launch sequence plays, new round begins, cycle returns to Phase 1	Watch → Full control

CYCLE INVARIANT

> Phase 4 (Emergency Systems) is not a discrete sequential phase — it runs **concurrently with Phase 3** and is triggered by in-game events. It is listed separately for documentation clarity. All other phases are strictly sequential.

1.2 Phase 1 — Launch Bay Briefing

The player begins each round inside the **launch tube** aboard the HumanSpaceBattleShip (model: **HumanSpaceBattleShip.glb**). The camera is positioned at the pilot's perspective — seated in the cockpit, facing the closed launch tube doors. The Launch Bay environment is dimly lit with warning lights and status displays visible on the tube walls.

The **Launch Bay Announcer** delivers the briefing via the ship's intercom. This is a distinct, authoritative voice character defined in **BFXG-005 (Voice Guide)** — commanding, calm, military cadence. The announcer is NOT the same voice as the Wingman or Command.

Briefing Delivery Sequence

1. **Threat Briefing** — What new enemy type, formation, or environmental hazard awaits this round. Example: "Sensors are detecting a Vex Swarm formation in Sector 7-Alpha. Expect fast-moving interceptors with cloaking capability."
2. **Mission Objective (Primary)** — The main goal the player must complete to finish the round. Types include: destroy target, escort convoy, clear sector, defeat boss, defend station, intercept bombers, disable capital ship.
3. **Secondary Objectives** — Optional bonus goals for extra XP and credits. Examples: destroy all enemy fighters before the timer expires, protect all convoy ships (zero losses), collect intel data pods.
4. **Weapon/Ability Inventory Check** — Current loadout is displayed on the HUD overlay within the cockpit. Announcer confirms: "Your loadout is confirmed: [WEAPON LIST]. All systems nominal."

5. **New Unlock Announcement** (if applicable) — If the player earned a new ability or weapon in the previous round's refit: "Pilot, you have been equipped with **[ABILITY/WEAPON NAME]**. Use it well."
6. **Rank Achievement Announcement** (if applicable) — If the player reached a new rank: "Congratulations, **[RANK NAME]**. Command expects great things."
7. **Clearance to Launch** — "You are cleared for launch. Good hunting, pilot."

> **DESIGN NOTE — ROUND 1 BRIEFING**
>
> The very first briefing (Round 1) includes additional tutorial language: identifying HUD elements, explaining controls, and noting the rescue shuttle system. Subsequent briefings omit tutorial elements unless a new mechanic is introduced.

1.3 Phase 2 — Launch Sequence

Upon clearance, the launch sequence cut scene plays:

8. Warning klaxon sounds — amber launch lights begin flashing along the tube walls
9. Launch tube doors slide open — deep space is visible ahead, stars and the combat zone in the distance
10. Electromagnetic catapult rails energize (visible blue energy along the tube floor)
11. The starfighter accelerates down the launch tube — G-force visual effects on cockpit camera
12. Fighter exits the carrier at high speed — camera pulls back to a chase-cam behind the fighter
13. The HumanSpaceBattleShip carrier recedes behind the player as they enter the combat zone

14. HUD activates with a power-on sweep animation — shields, fuel, weapons, radar all come online
15. Player is given full flight control

Concept Reference — HUD activation sequence as the pilot enters the combat zone

1.4 Phase 3 — Active Combat / Mission

This is the core gameplay phase. The player has full control of their starfighter and must complete the primary mission objective to advance. All HUD elements are active. Enemy AI behaviors follow **BFXG-002 (Starfighter ANPC Guideline)**.

Mid-Mission Announcements

During active combat, two voice sources provide information to the player:

Voice Source	Character	Callout Types
Command	Ship's tactical officer (calm, professional)	Enemy wave incoming alerts, objective progress updates ("3 of 5 targets destroyed"), boss appearance warning (Round 3+), mission timer warnings
Wingman	Fellow pilot (energetic, personal)	Tactical callouts ("Break left! Bogey on your six!"), encouragement ("Nice shot!"), warnings ("Watch your fuel, lead"),

Voice Source	Character	Callout Types
		situational awareness ("I see two more coming from the asteroid field")

HUD Elements During Combat

- **Shield Meter** — Regenerating energy shield around the hull
- **Hull Integrity Meter** — Physical damage to the ship (does NOT regenerate)
- **Fuel Gauge** — Depletes over time and with afterburner use
- **Ammo / Cooldown Indicators** — Per weapon, showing available rounds or cooldown timers
- **Radar / Minimap** — Enemy positions, objective markers, allied ships
- **Mission Objective Tracker** — Current primary and secondary objective status
- **Rescue Shuttle Tokens** — 3 shuttle icons showing remaining lives
- **Countermeasure Count** — Remaining flares/chaff

1.5 Phase 4 — In-Combat Emergency Systems

Three critical systems operate during active combat. These are lifeline mechanics designed to extend gameplay and create dramatic tension.

A. RESCUE SHUTTLE SYSTEM — 3 LIVES PER GAME

CRITICAL GAME VALUE

The player receives exactly **3 Rescue Shuttle uses per ENTIRE GAME** — NOT per round. Once all 3 are consumed, the next ship disable triggers **GAME OVER**.

Trigger Condition: The player's hull integrity reaches 0. The ship does **NOT** explode immediately.

Rescue Sequence:

16. Ship goes dead — power flickers, controls unresponsive, ship drifts
17. Announcer: "Hull integrity critical. Rescue shuttle dispatched. Stand by for emergency repairs."
18. A **Rescue Shuttle** (model: **Rescue Shuttle.glb**) arrives — visible approaching from the carrier's direction
19. Cut scene (10 seconds): Rescue Shuttle docks with the disabled fighter using a tractor beam
20. Animated repair sequence — sparks, welding effects, hull plating being replaced
21. Ship systems reboot — HUD comes back online sequentially
22. Ship is fully repaired: hull restored to 100%, shields recharged, ammo refilled, fuel topped off
23. Player is repositioned at a safe re-entry point away from immediate enemy fire
24. Full control is returned to the player

HUD Indicator: Three shuttle icons displayed in the bottom-left corner of the HUD. Icons are **bright white** when available and **greyed out with a red X** when consumed. When the final shuttle is used, an alarm sounds and the announcer warns: "That was your last rescue shuttle, pilot. There will be no more."

B. FUEL SYSTEM

&

EMERGENCY REFUELING

Fuel is a persistent resource that depletes during flight and creates strategic pressure.

Fuel Depletion Rules

Activity	Fuel Consumption Rate
Normal flight	Standard depletion (baseline)
Afterburner / Boost	**3x baseline** consumption rate
Idle / Drifting	0.25x baseline (life support only)

Fuel Gauge Color Coding

Fuel Level	Gauge Color	Alert Status
75% – 100%	**GREEN**	Nominal — no alerts
25% – 74%	**YELLOW**	Caution — monitor fuel usage
1% – 24%	**RED (FLASHING)**	Alarm sounds, HUD flashes red warning, announcer: "Fuel critical."
0%	**EMPTY — DEAD STICK**	Ship drifts, no thrust. Weapons still operate. 30-second countdown begins.

Emergency Refueling Procedure

> **CRITICAL GAME VALUE**
>
> Emergency refueling is limited to **2 calls per round**. The RED ROUND FUEL BUTTON is a prominent, always-visible UI element near the fuel gauge.

25. Player presses the **RED ROUND FUEL BUTTON** on the HUD (or mapped controller/keyboard input)

26. Announcer: "Fuel tanker inbound. Hold position for refueling."
27. A **Fuel Tanker** ship arrives and approaches the player's fighter
28. The tanker docks mid-flight — a fuel line extends and connects
29. Player must hold **relatively steady** for **5 seconds** during refueling (minor movement allowed, but sharp turns break the connection and restart the 5-second timer)
30. Fuel gauge refills to 100%
31. Tanker detaches and departs
32. Remaining refueling calls indicator decrements

Dead Stick Scenario: If fuel reaches 0% and the player has no refueling calls remaining, a **30-second countdown** begins. If the countdown expires, a Rescue Shuttle is auto-dispatched (consuming one rescue shuttle token). If no rescue shuttle tokens remain, the result is **GAME OVER**.

C. COUNTERMEASURES & MISSILE WARNING

33. Enemy missile locks onto the player — targeting alarm sounds
34. HUD displays: **"INCOMING MISSILE — DEPLOY COUNTERMEASURES"** with a directional indicator showing the missile's approach vector
35. Player deploys flares/chaff (limited supply — replenished only at dock between rounds)
36. If countermeasures succeed: missile is diverted, no damage
37. If countermeasures fail or are not deployed: missile impacts shield first, then hull if shields are down

Countermeasure capacity is displayed near the weapon loadout area of the HUD. Starting capacity and replenishment amounts scale with progression (see BFXG-008).

1.6 Phase 5 — Mission Complete

When the primary objective is achieved (all enemies destroyed, boss defeated, escort arrived safely, etc.), the mission complete sequence triggers:

38. **Announcement:** Announcer OR Wingman declares: "Mission complete! Outstanding work, pilot. Autopilot engaged — returning to base."
39. A brief **victory fanfare** plays (musical sting, 3-4 seconds)
40. Player controls are disabled — the ship enters **autopilot mode**
41. The HUD displays the **AUTOPILOT** indicator (amber text, center screen)
42. The ship banks and turns toward the HumanSpaceBattleShip carrier, visible in the distance
43. The return flight is a **relaxing cinematic moment** — the camera shows the space environment, passing debris from the battle, other allied ships also returning, nebulae or planetary bodies in the background
44. Player **cannot control** the ship during autopilot return but can look around (camera rotation only)
45. Duration: approximately **15–20 seconds** of return flight before docking sequence begins

1.7 Phase 6 — Docking Cut Scene

The docking sequence is a detailed cinematic that reinforces the scale and physicality of the carrier. It transitions the player from the space environment back to the interior launch bay.

Docking Sequence Steps

Step	Action	Camera	Duration
1	Ship approaches carrier from outside — carrier grows larger in view	External chase cam behind fighter	4 sec
2	Landing lights on the carrier become visible, guiding the approach	Chase cam, slight angle shift	3 sec
3	Ship aligns with docking bay entrance — alignment HUD appears	Cockpit interior (pilot POV)	3 sec
4	Bay doors open, guide lights activate (green strips along the bay entrance)	Cockpit interior looking forward	3 sec
5	Ship enters docking bay — interior visible (other fighters parked, crew working, equipment, tool racks)	External side cam inside the bay	4 sec
6	Ship settles onto landing pad, landing gear deploys with hydraulic sounds	Low angle from bay floor	3 sec
7	Docking clamps engage — mechanical locking sounds, ship powers down, engines wind down	Close-up on clamp mechanisms	2 sec
8	Camera transitions to interior cockpit — pilot's perspective inside the launch bay	Cockpit interior, ambient bay sounds	2 sec

Step	Action	Camera	Duration
9	Ship is moved via rail system back to the launch tube position — mechanical rail movement visible and audible	Side tracking cam following the rail	4 sec

Total docking cut scene duration: approximately **28 seconds**. The cut scene is **skippable** after the first playthrough (Round 1 docking is forced/unskippable for narrative immersion).

1.8 Phase 7 — Refit & Reward

The player is now back in the launch bay — this serves as the **home screen between rounds**. The camera shows the cockpit interior with the launch tube doors closed ahead, and refit activity visible through side viewports and status displays.

Debrief & Refit Sequence

46. **Welcome Back:** Announcer: "Welcome back, pilot. Your fighter is being refueled, repaired, and reprovisioned."
47. **Animated Refit Sequence** (visible through cockpit viewports and on status screens):
    - Fuel lines connecting and gauge refilling
    - Repair drones working on hull damage — sparks, plating replacement
    - Ammo canisters being loaded into weapon bays
    - Countermeasure pods being restocked
48. **Performance Summary** — displayed on the cockpit's center MFD (multi-function display):

Metric	Display Format
Enemies Destroyed	Count (e.g., "14 hostiles eliminated")
Accuracy Percentage	Percentage (e.g., "72% accuracy")
Primary Objective	COMPLETED / FAILED
Secondary Objectives	Count completed / total (e.g., "2 of 3")
Time Elapsed	Minutes:Seconds (e.g., "04:32")
XP Earned	Numeric with progress bar toward next rank
Credits Earned	Numeric (running total displayed)
Rank Progress	Progress bar showing current position toward next rank
Rescue Shuttles Remaining	Icon display (e.g., 2 of 3 remaining)

Reward Announcements

- **New Ability/Weapon:** "You have earned **[ABILITY/WEAPON NAME]**." — displayed with description text and an icon/3D preview of the item. The new item is automatically added to the player's loadout.
- **Rank Promotion:** "You have achieved the rank of **[RANK NAME]**." — displayed with the rank insignia. If a rank threshold is crossed, a **full rank ceremony animation** plays: insignia animates onto the cockpit console, a brief trumpet fanfare sounds, and the new rank title pulses on screen for 3 seconds.

1.9 Phase 8 — Test Flight (Optional)

After rewards are announced, a prominent UI button appears on the cockpit display: **"TEST FLIGHT"**. This is entirely optional — the player can skip directly to the next mission briefing.

Test Environment Specifications

Element	Description
Environment	Open space area near the carrier, no active enemies, safe zone
Stationary Target Dummies	Non-moving targets for weapon testing — display damage numbers on hit
Moving Target Drones	AI drones that fly in patterns for tracking/aiming practice
Obstacle Course	Asteroid or debris field with marker gates for maneuverability testing
Ammo & Fuel	**Unlimited** — test mode does not consume real resources
Time Limit	**None** — player can fly as long as they want

Test Flight Controls

During the test flight, two buttons are available on the HUD:

- **"RETURN TO LAUNCH BAY"** — Engages autopilot, flies back to the carrier, returns to the dock screen. Player then proceeds to the next mission briefing.
- **"SKIP TO MISSION"** — Immediately ends the test flight and transitions directly to the next mission briefing (Phase 9). No return flight animation.

1.10 Phase 9 — Next Mission Briefing

When the player is ready (either skipped test flight, returned from test flight, or dismissed the test flight prompt), the announcer delivers the next round's briefing using the same format as Phase 1. Content is updated for the new round: new threats, new objectives, updated loadout display reflecting any newly acquired abilities or weapons.

The briefing concludes: "You are cleared to launch. Good hunting, pilot."

1.11 Phase 10 — Launch / Cycle Repeats

The launch sequence plays again (Phase 2), and the new round begins. The cycle continues until the player triggers a Game Over condition or completes the final round (Round 20).

2. Game Over Conditions

ID	Condition	Trigger	Result
GO-1	**All Rescue Shuttles Exhausted**	Player has used all **3 rescue shuttles** AND hull integrity reaches 0 again	Ship explodes. Game Over cut scene plays. Final score displayed.
GO-2	**Fuel Death Spiral**	Fuel reaches 0%, no refueling calls remain, no rescue shuttle tokens remain, 30-second countdown expires	Ship goes permanently dead stick. Captured/destroyed. Game Over.
GO-3	**Voluntary Ejection**	Player chooses to eject (mercy mechanic). Available from the pause menu at any time during combat.	Pilot ejects. Rank progress is **saved**. Credits are saved. Current run ends. Player returns to main menu with progress intact for next run.
GO-4	**Victory — Final Mission Complete**	Player completes Round 20's primary objective	Victory cut scene. Final rank and score displayed. Credits roll. Game Complete.

DESIGN NOTE — EJECTION MERCY MECHANIC

The voluntary ejection exists to prevent player frustration. If a player is on their last rescue shuttle with low fuel in a difficult round, they can eject to preserve their rank and credits rather than losing everything. This encourages players to attempt difficult rounds knowing they have a safety net. The eject option is accessed via the pause menu — it is **not** a one-button action to prevent accidental use.

3. Launch Bay Announcer Script Templates

All announcer lines are parameterized. Variables are shown in **{BRACKETS}**. The game engine selects randomly from available variations each round to prevent repetition. Voice profile is defined in **BFXG-005**.

3.1 Pre-Launch Briefing

> Variation A:
>
> "Attention pilot. Intelligence reports
>
> **{THREAT_DESCRIPTION}**
>
> in
>
> **{SECTOR_NAME}**
>
> . Your primary objective:
>
> **{PRIMARY_OBJECTIVE}**
>
> . Secondary targets include
>
> **{SECONDARY_OBJECTIVES}**
>
> . Your loadout is confirmed. You are cleared for launch. Good hunting, pilot."

> Variation B:
>
> "Pilot, listen up. We have

{THREAT_DESCRIPTION}

bearing

{BEARING}

. Command needs you to

{PRIMARY_OBJECTIVE}

. Bonus targets are marked on your nav computer:

{SECONDARY_OBJECTIVES}

. All systems green. Launch when ready."

Variation C:

"Stand by for mission briefing. Sensor sweep shows

{THREAT_DESCRIPTION}

across

{SECTOR_NAME}

. Mission priority:

{PRIMARY_OBJECTIVE}

. If opportunity allows,

{SECONDARY_OBJECTIVES}

. Weapons hot, systems nominal. Tube is clear — launch, launch, launch."

3.2 Mid-Mission Callouts

Enemy Wave Incoming

> A:
>
> "New contacts on scope.
>
> **{ENEMY_COUNT}**
>
> **{ENEMY_TYPE}**
>
> inbound, bearing
>
> **{BEARING}**
>
> ."

> B:
>
> "Heads up — additional hostiles detected.
>
> **{ENEMY_COUNT}**
>
> **{ENEMY_TYPE}**
>
> entering the combat zone."

> C:
>
> "Warning: enemy reinforcements.

> **{ENEMY_COUNT}**
>
> targets, classification
>
> **{ENEMY_TYPE}**
>
> . Engage at will."

Objective Progress

> A:
>
> "
>
> **{COMPLETED_COUNT}**
>
> of
>
> **{TOTAL_COUNT}**
>
> targets neutralized. Keep it up, pilot."

> B:
>
> "Objective update:
>
> **{COMPLETED_COUNT}**
>
> down,
>
> **{REMAINING_COUNT}**
>
> to go."

> C:
>
> "Progress report —
>
> **{COMPLETED_COUNT}**
>
> of
>
> **{TOTAL_COUNT}**
>
> confirmed kills. Finish them off."

Boss Appearance Warning (Round 3+)

> A:
>
> "Massive contact emerging. That's a
>
> **{BOSS_NAME}**
>
> . All hands, priority target!"

> B:
>
> "Command to pilot — high-value target detected:
>
> **{BOSS_NAME}**
>
> . Focus all firepower."

> C:
>
> "Pilot, be advised:
>
> **{BOSS_NAME}**
>
> just dropped out of warp. That's your primary. Take it down."

3.3 Mission Complete

> A:
>
> "Mission complete! Outstanding work, pilot. Autopilot engaged — returning to base."

> B:
>
> "All objectives achieved. Excellent flying out there. Setting autopilot for home. Sit back and enjoy the ride."

> C:
>
> "Sector clear, mission accomplished. You're heading home, pilot. Autopilot

> engaged."

3.4 Docking Welcome Back

> A:
>
> "Welcome back, pilot. Your fighter is being refueled, repaired, and reprovisioned."

> B:
>
> "Good to have you back in one piece. Maintenance crews are on your bird now."

> C:
>
> "Docking confirmed. Refit operations commencing. Stand by for debrief."

3.5 Refit Status

> A:
>
> "Fuel tanks topped off. Ammo loaded. Hull integrity restored. Your ship is mission-ready."

> B:
>
> "All systems green — fuel, weapons, shields, hull. You're good to go when you're ready."

> C:
>
> "Refit complete. Your fighter is in top condition. Awaiting your orders, pilot."

3.6 Reward Announcement

New Ability / Weapon

> A:
>
> "Pilot, you have been equipped with
>
> **{REWARD_NAME}**
>
> .

> **{REWARD_DESCRIPTION}**
>
> . Use it well."

> B:
>
> "Armory has authorized a new addition to your loadout:
>
> **{REWARD_NAME}**
>
> .
>
> **{REWARD_DESCRIPTION}**
>
> ."

> C:
>
> "Good news — you've unlocked
>
> **{REWARD_NAME}**
>
> .
>
> **{REWARD_DESCRIPTION}**
>
> . It's already installed."

Rank Promotion

> A:
>
> "Attention on deck. You have achieved the rank of
>
> **{RANK_NAME}**
>
> . Command salutes your service."

> B:
>
> "Congratulations,
>
> **{RANK_NAME}**
>
> . You've earned it. Command expects great things."

> C:
>
> "By order of Fleet Command, you are hereby promoted to
>
> **{RANK_NAME}**
>
> . Wear it with pride."

3.7 Test Flight Invitation

> A:

> "A test flight area is available if you'd like to try out your new gear. Launch when ready, or proceed to your next mission."

> B:
>
> "Want to take your new equipment for a spin? The test range is clear. No pressure — take your time."

> C:
>
> "Test flight option is available. Practice area is prepped with targets and drones. Your call, pilot."

3.8 Next Mission Briefing

> A:
>
> "Next mission is ready.
>
> **{THREAT_DESCRIPTION}**
>
> detected in

> **{SECTOR_NAME}**
>
> . Objective:
>
> **{PRIMARY_OBJECTIVE}**
>
> . You are cleared to launch."

> B:
>
> "Pilot, your next sortie is queued.
>
> **{THREAT_DESCRIPTION}**
>
> are massing at
>
> **{SECTOR_NAME}**
>
> . Mission:
>
> **{PRIMARY_OBJECTIVE}**
>
> . Good hunting."

> C:
>
> "No rest for the brave. We've got
>
> **{THREAT_DESCRIPTION}**
>
> in
>
> **{SECTOR_NAME}**

> . You know the drill —
>
> **{PRIMARY_OBJECTIVE}**
>
> . Launch when ready."

3.9 Rescue Shuttle Dispatch

> A:
>
> "Hull integrity critical. Rescue shuttle dispatched. Stand by for emergency repairs."

> B:
>
> "Mayday received. Rescue shuttle is en route. Hold tight, pilot — help is coming."

> C:
>
> "You're hit bad. Emergency rescue inbound. Activating tractor beam recovery."

> Last Shuttle Warning:
>
> "That was your last rescue shuttle, pilot. There will be no more. Fly careful."

3.10 Fuel Tanker Dispatch

> A:
>
> "Fuel tanker inbound. Hold position for refueling."

> B:
>
> "Copy your fuel request. Tanker is on approach — maintain a steady heading."

> C:
>
> "Tanker dispatched to your coordinates. Hold still for five seconds during fuel transfer."

> Last Refuel Warning:

> "That was your last refueling call this round. Manage your fuel, pilot."

3.11 Game Over

> A (Destruction):
>
> "We've lost the signal. All rescue assets expended. Pilot down.
>
> [Pause]
>
> They flew with honor."

> B (Destruction):
>
> "Contact lost. No rescue available.
>
> [Pause]
>
> Flight log saved. Their sacrifice will be remembered."

> C (Ejection):
>
> "Ejection confirmed. Pilot recovered safely. Mission aborted — rank and credits preserved. Live to fly another day."

4. HUD Elements Reference

ID	Element	Screen Position	Display Details	Color Coding
H-01	**Shield Meter**	Top Left	Horizontal bar showing shield energy level (0–100%). Regenerates over time when not taking damage.	Blue (#0066cc) fill, dims as depleted
H-02	**Hull Integrity Meter**	Top Left (below shields)	Horizontal bar showing structural hull health (0–100%). Does NOT regenerate.	Green >50%, Yellow 25–50%, Red <25%
H-03	**Fuel Gauge**	Top Right	Vertical bar or arc gauge showing fuel remaining (0–100%). Depletes over time and with boost.	Green 75–100%, Yellow 25–74%, Red flashing <25%
H-04	**Fuel Refuel Calls Remaining**	Top Right (near fuel gauge)	Small numeric indicator: "REFUEL: **{X}**/2" — shows remaining emergency refueling calls this round.	White text; RED ROUND FUEL BUTTON pulses when fuel <25%
H-05	**Weapon Loadout**	Bottom Center	Icons for each equipped weapon/ability. Shows ammo count or cooldown timer per weapon. Active weapon	Active: white/bright; Cooldown: amber; Empty: greyed out

ID	Element	Screen Position	Display Details	Color Coding
			highlighted.	
H-06	**Radar / Minimap**	Bottom Right	Circular radar showing: red dots (enemies), green dots (allies), yellow diamond (objective), white triangle (player facing direction).	Red: enemy, Green: ally, Yellow: objective
H-07	**Mission Objective Tracker**	Top Center	Text display: primary objective name + progress (e.g., "Destroy Bombers: 3/5"). Secondary objectives listed below in smaller text.	White text; completed items turn green with checkmark
H-08	**Rescue Shuttle Tokens**	Bottom Left	3 shuttle icons representing remaining rescue uses. Icons are bright when available, greyed with red X when consumed.	Available: white; Used: grey with red X
H-09	**Countermeasure Count**	Bottom Center (near weapons)	Numeric count of remaining flares/chaff. Icon of a flare burst next to the number.	White; flashes amber when ≤ 2 remaining
H-10	**Speed Indicator**	Left Center	Numeric readout of current speed. Changes with thrust input and	White; turns blue during afterburner

ID	Element	Screen Position	Display Details	Color Coding
			afterburner activation.	
H-11	**Targeting Reticle**	Center Screen	Crosshair that indicates weapon aim point. Tightens when targeting an enemy (lock-on).	Green: neutral; Red: enemy targeted; flashing red: missile lock
H-12	**Enemy Lock Indicator**	Center Screen (around reticle)	Diamond bracket that narrows and locks around an enemy when within weapon range and aligned.	Yellow: tracking; Red: locked
H-13	**Incoming Missile Warning**	Center Screen (overlay)	Large flashing text: "INCOMING MISSILE" with directional arrow showing missile approach vector. Audio alarm accompanies.	Red flashing text and border
H-14	**Rank / XP Bar**	Top Right Corner	Subtle thin bar showing XP progress toward next rank. Current rank abbreviation displayed. Non-intrusive during combat.	Gold fill on dark background
H-15	**Score / Credits Counter**	Top Right (below rank bar)	Running total of credits earned this run. Increments with each kill	White text; pulses gold briefly on increment

ID	Element	Screen Position	Display Details	Color Coding
			and objective completion.	
H-16	**Round Number Indicator**	Top Center (left of objective)	Displays: "ROUND **{X}** / 20"	White text
H-17	**Boss Health Bar**	Top Center (full width)	Appears ONLY when a boss is engaged. Large bar with boss name. Shows health segments for multi-phase bosses.	Red fill; segments separated by white dividers
H-18	**Autopilot Indicator**	Center Screen	Appears during return flight (Phase 5). Displays: "AUTOPILOT ENGAGED" — controls disabled.	Amber text, steady (not flashing)

5. Cut Scene Specifications

5.1 Launch Sequence

Property	Specification
Duration	8–10 seconds
Camera Angles	Starts in cockpit POV looking down the launch tube. Transitions to external chase cam as fighter exits. Wide shot as carrier recedes.
Audio — Music	Rising orchestral swell, transition to combat music as HUD activates
Audio — SFX	Klaxon alarm, catapult rail charging hum, acceleration roar, launch tube

Property	Specification
	echo transitioning to open space silence, then engine hum
Audio — Voice	Announcer clearance line plays before cut scene begins
Player Interaction	Skippable after Round 1 (hold SKIP button for 1 second to prevent accidental skips)
Transitions	Fade from briefing UI → Launch tube interior → Space exterior → HUD activation → Player control

5.2 Rescue Shuttle Arrival & Repair

Property	Specification
Duration	**10 seconds** (fixed — not skippable, provides dramatic tension)
Camera Angles	External: disabled fighter drifting. Rescue Shuttle approaches from above/behind. Close-up of tractor beam locking on. Side view of docking. Close-up of repair sparks. Pull-back as systems reboot.
Audio — Music	Tense, low strings during approach. Resolves to hopeful brass as repair completes.
Audio — SFX	Ship power failure sounds (flickering, static), tractor beam hum, metal-on-metal docking clank, welding/sparking, system reboot beeps
Audio — Voice	Rescue Shuttle dispatch line at start; "Repairs complete. Get back out there." at end.
Player Interaction	**Not skippable.** Player watches the full 10-second sequence. This is intentional — the cost of losing a life is time and tension.
Transitions	Combat freeze → External camera → Repair animation → Cockpit reboot → Reposition → Player control

5.3 Fuel Tanker Docking

Property	Specification
Duration	5 seconds (the refueling hold period) — player retains partial control
Camera Angles	Remains in gameplay camera. Tanker approaches from behind/above (visible in cockpit mirrors or periphery). Fuel line extends — visible on the HUD viewport. No full cut scene — gameplay continues.
Audio — SFX	Tanker engine approach rumble, fuel line connection clunk, liquid transfer sound, disconnection pop
Audio — Voice	Tanker dispatch line. "Refueling complete. Tanker disengaging." on finish.
Player Interaction	Player must hold relatively steady. Sharp turns break the fuel line — 5-second timer resets. Player CAN still fire weapons and deploy countermeasures during refueling.
Transitions	None — seamless in-gameplay event

5.4 Mission Complete / Autopilot Return

Property	Specification
Duration	15–20 seconds
Camera Angles	Player camera with rotation freedom (look around). Chase cam option. Scenic shots of the space environment — battle debris, allied ships, celestial bodies.
Audio — Music	Triumphant but calming orchestral — decompression from combat intensity
Audio — SFX	Engine cruising hum, occasional radio

Property	Specification
	chatter from other ships, carrier approach beacons
Audio — Voice	Mission complete announcement at start. Optional Wingman chatter: "That was one for the books." / "Drinks are on me when we dock."
Player Interaction	Camera look only — no flight control. Not skippable (transitions seamlessly into docking).
Transitions	Seamless transition into Phase 6 docking cut scene as carrier fills the viewport

5.5 Docking Approach & Landing

Property	Specification
Duration	~28 seconds (see Phase 6 breakdown above)
Camera Angles	9-step camera sequence detailed in Phase 6 table — alternates between external chase cam, cockpit POV, bay interior side cam, low-angle floor cam, close-ups, and tracking cam.
Audio — Music	Low, ambient military theme — sense of homecoming
Audio — SFX	Approach thrusters, bay door hydraulics, guide light pings, landing gear deployment, docking clamp locks, engine power-down whine, rail system mechanical movement
Audio — Voice	Docking control: "Approach confirmed. Bay doors opening." / "Docking clamps engaged. Welcome home."
Player Interaction	Skippable after Round 1 (hold SKIP for 1 second)
Transitions	External space → Bay interior → Cockpit interior → Launch tube position → Refit UI overlay

5.6 Interior Launch Bay — Refit Sequence

Property	Specification
Duration	12–15 seconds for the animated refit; performance summary remains until player dismisses
Camera Angles	Cockpit POV — looking out through canopy at refit activity. Side viewport shows repair drones, fuel lines, ammo loading. Status screens on cockpit console display performance data.
Audio — Music	Quiet, mechanical ambient — workshop atmosphere with distant radio chatter
Audio — SFX	Fuel pump hiss, welding sparks, ammo magazine clicks, hydraulic arm movements, system diagnostic beeps
Audio — Voice	Welcome back line, then refit status line, then reward announcements (if any), then test flight invitation.
Player Interaction	Player reviews performance summary. Dismisses with button press. Interacts with Test Flight or Next Mission buttons.
Transitions	Refit animation → Performance overlay → Reward animation (if applicable) → Test Flight prompt or Next Briefing

5.7 Rank Promotion Ceremony

Property	Specification
Duration	5–7 seconds (plays during the refit phase when a rank is earned)
Camera Angles	Close-up on cockpit console where rank insignia is displayed. Insignia animates — old rank fades, new rank materializes with a golden particle effect.
Audio — Music	Brief trumpet/brass fanfare (3–4 seconds)
Audio — SFX	Ceremonial chime, insignia stamp/seal

Property	Specification
	sound
Audio — Voice	Rank promotion announcement line
Player Interaction	Not skippable — brief and rewarding. Auto-dismisses after the fanfare.
Transitions	Overlays on top of the refit phase UI, then returns to normal refit view

5.8 Test Flight Launch & Return

Property	Specification
Launch Duration	5 seconds (abbreviated launch — no full briefing, shorter tube exit)
Return Duration	8 seconds (abbreviated docking — faster approach, skip interior details)
Camera	Same as standard launch/dock but abbreviated cuts — fewer camera angles, shorter holds
Audio	Relaxed version of launch/dock audio — no klaxon, lighter music
Player Interaction	Both launch and return are skippable immediately (no Round 1 restriction)
Transitions	Launch Bay → Test Area (with "TEST MODE" watermark on HUD) → Launch Bay on return

5.9 Game Over

Property	Specification
Duration — Destruction	8–10 seconds
Duration — Ejection	6 seconds
Camera — Destruction	Cockpit POV: instruments fail, cracks appear in canopy, flash of light. External: ship detonates in a fireball.

Property	Specification
	Camera pulls back slowly as debris drifts. Fade to black.
Camera — Ejection	Cockpit POV: canopy jettisons, seat rockets upward. External: ejection pod tumbles away from the fighter. Rescue beacon activates (blinking light). Fade to white.
Audio — Destruction	System failure alarms, explosion, silence, then somber orchestral
Audio — Ejection	Canopy explosive bolts, ejection rocket whoosh, beacon ping, hopeful but melancholic theme
Audio — Voice	Game Over announcer line (see Section 3.11)
Player Interaction	Not skippable. After fade, Game Over screen displays final score, rank achieved, rounds completed. "PLAY AGAIN" and "MAIN MENU" buttons.
Transitions	Gameplay → Destruction/Ejection animation → Fade → Game Over score screen

6. Asset Reference Table

6.1 3D Models

ID	Asset Name	File / Reference	Used In Phases
M-01	**HumanSpaceBattleShip** (Home Base Carrier)	HumanSpaceBattleShip.glb	1, 2, 5, 6, 7, 8, 9, 10
M-02	**Player Starfighter**	Per BFXG-008 (variant based on progression)	All phases
M-03	**Rescue Shuttle**	Rescue Shuttle.glb	4A (Rescue)
M-04	**Fuel Tanker Ship**	FuelTanker.glb	4B (Refueling)

ID	Asset Name	File / Reference	Used In Phases
M-05	**Target Dummies** (Stationary)	TestTarget_Static.glb	8 (Test Flight)
M-06	**Target Drones** (Moving)	TestTarget_Drone.glb	8 (Test Flight)
M-07	**Launch Bay Interior** (Environment)	LaunchBay_Interior.glb	1, 7, 9
M-08	**Docking Bay Interior** (Environment)	DockingBay_Interior.glb	6
M-09	**Repair Drones**	RepairDrone.glb	7 (Refit animation)
M-10	**Ejection Pod**	EjectionPod.glb	Game Over (ejection)
M-11	**Obstacle Course Gates**	TestCourse_Gate.glb	8 (Test Flight)

6.2 UI Elements & HUD Sprites

ID	Asset	Description
U-01	Shield Meter Sprite	Horizontal bar with blue fill gradient and border
U-02	Hull Integrity Meter Sprite	Horizontal bar with green/yellow/red fill states
U-03	Fuel Gauge Sprite	Vertical bar or arc with green/yellow/red states
U-04	RED ROUND FUEL BUTTON	Prominent circular button — bright red with fuel icon, pulses when fuel is low
U-05	Weapon Loadout Icons	Icon set for each weapon/ability type (per BFXG-008 unlock list)
U-06	Radar Display	Circular radar background with sweep line animation
U-07	Rescue Shuttle Token	Shuttle silhouette —

ID	Asset	Description
	Icons (x3)	white (available) / grey with red X (used)
U-08	Targeting Reticle Set	Crosshair in neutral (green), targeting (yellow), locked (red) states
U-09	Missile Warning Overlay	Red flashing border and directional arrow graphics
U-10	Rank Insignia Set	Icon for each rank tier (per BFXG-008 rank table)
U-11	Autopilot Indicator	Amber "AUTOPILOT ENGAGED" text overlay
U-12	Boss Health Bar	Full-width bar with segmented fill and boss name text area
U-13	Test Flight Button	"TEST FLIGHT" button — cockpit UI style
U-14	Return to Launch Bay Button	"RETURN TO LAUNCH BAY" button — cockpit UI style
U-15	Countermeasure Icon	Flare burst silhouette with numeric counter

6.3 Sound Effects & Music

ID	Asset	Used In
S-01	Launch klaxon alarm	Phase 2 — Launch
S-02	Catapult rail charge & acceleration	Phase 2 — Launch
S-03	HUD power-on sweep	Phase 2 — HUD activation
S-04	Missile lock alarm	Phase 4C — Incoming missile
S-05	Countermeasure deploy (flare pop)	Phase 4C

ID	Asset	Used In
S-06	Tractor beam hum	Phase 4A — Rescue shuttle
S-07	Welding / repair sparks	Phase 4A — Rescue, Phase 7 — Refit
S-08	Fuel transfer liquid sound	Phase 4B — Refueling
S-09	Victory fanfare (musical sting)	Phase 5 — Mission complete
S-10	Docking bay door hydraulics	Phase 6 — Docking
S-11	Landing gear deployment	Phase 6 — Docking
S-12	Docking clamp engagement	Phase 6 — Docking
S-13	Rail system mechanical movement	Phase 6 — Docking
S-14	Rank promotion fanfare (brass)	Phase 7 — Rank ceremony
S-15	Explosion (ship destruction)	Game Over — Destruction
S-16	Ejection seat rocket	Game Over — Ejection
S-17	Rescue beacon ping	Game Over — Ejection
S-18	Low fuel alarm	Phase 4B — Fuel critical
S-19	System reboot sequence beeps	Phase 4A — Post-rescue
S-20	Engine cruising hum (ambient)	Phase 3, 5 — Flight

6.4 Voice Line Categories

ID	Category	Min. Variations	Voice Actor
V-01	Pre-Launch Briefing	3	Launch Bay Announcer
V-02	Enemy Wave Alert	3	Command
V-03	Objective Progress	3	Command

ID	Category	Min. Variations	Voice Actor
V-04	Boss Warning	3	Command
V-05	Mission Complete	3	Announcer / Wingman
V-06	Docking Welcome	3	Announcer
V-07	Refit Status	3	Announcer
V-08	Reward — Weapon/Ability	3	Announcer
V-09	Reward — Rank	3	Announcer
V-10	Test Flight Invitation	3	Announcer
V-11	Next Mission Briefing	3	Announcer
V-12	Rescue Shuttle Dispatch	3 + 1 final warning	Announcer
V-13	Fuel Tanker Dispatch	3 + 1 final warning	Announcer
V-14	Game Over	3 (2 destruction + 1 ejection)	Announcer
V-15	Wingman Tactical Callouts	10+	Wingman
V-16	Wingman Encouragement	5+	Wingman
V-17	Wingman Warnings	5+	Wingman

7. Game State Machine

7.1 State Definitions

ID	State	Description
S01	**LAUNCH_BAY_BRIEFING**	Player in launch tube, announcer delivering briefing. No player flight

ID	State	Description
		control.
S02	**LAUNCH_SEQUENCE**	Launch cut scene playing. Transition from launch bay to space.
S03	**ACTIVE_COMBAT**	Player has full flight control. Enemies active. Objectives tracked.
S04	**RESCUE_IN_PROGRESS**	Player disabled. Rescue shuttle 10-second cut scene playing. No player control.
S05	**REFUELING_IN_PROGRESS**	Fuel tanker docked. 5-second hold timer active. Player has partial control.
S06	**MISSION_COMPLETE_AUTOPILOT**	All objectives met. Ship on autopilot returning to carrier. Camera-look only.
S07	**DOCKING_CUTSCENE**	Docking sequence cinematic. No player control.
S08	**REFIT_AND_REWARD**	Launch bay home screen. Performance summary, rewards, refit animation. UI interaction.
S09	**TEST_FLIGHT**	Optional practice environment. Full flight control. No enemies. Unlimited resources.
S10	**NEXT_MISSION_BRIEFING**	Announcer delivers next round briefing. Identical structure to S01 but with updated data.
S11	**GAME_OVER**	Terminal state. Destruction or ejection cut scene, then score screen.

7.2 State Transition Table

From State	To State	Trigger Condition
LAUNCH_BAY_BRIEFING	LAUNCH_SEQUENCE	Briefing complete, clearance given
LAUNCH_SEQUENCE	ACTIVE_COMBAT	Launch cut scene ends, HUD activates
ACTIVE_COMBAT	RESCUE_IN_PROGRESS	Hull integrity reaches 0 AND rescue shuttles > 0
ACTIVE_COMBAT	REFUELING_IN_PROGRESS	Player presses RED FUEL BUTTON AND refuel calls > 0
ACTIVE_COMBAT	MISSION_COMPLETE_AUTOPILOT	Primary objective completed
ACTIVE_COMBAT	GAME_OVER	Hull = 0 AND rescue shuttles = 0, OR fuel countdown expires AND rescue shuttles = 0, OR player ejects
RESCUE_IN_PROGRESS	ACTIVE_COMBAT	10-second repair sequence complete, ship repositioned
REFUELING_IN_PROGRESS	ACTIVE_COMBAT	5-second hold complete, fuel refilled
MISSION_COMPLETE_AUTOPILOT	DOCKING_CUTSCENE	Return flight reaches carrier proximity
DOCKING_CUTSCENE	REFIT_AND_REWARD	Docking sequence complete, ship in launch tube
REFIT_AND_REWARD	TEST_FLIGHT	Player presses "TEST FLIGHT" button
REFIT_AND_REWARD	NEXT_MISSION_BRIEFING	Player skips test flight or dismisses refit screen
TEST_FLIGHT	REFIT_AND_REWARD	Player presses "RETURN TO LAUNCH BAY"
TEST_FLIGHT	NEXT_MISSION_BRIEFING	Player presses "SKIP TO MISSION"
NEXT_MISSION_BRIEFING	LAUNCH_SEQUENCE	Briefing complete, clearance given
ACTIVE_COMBAT	RESCUE_IN_PROGRESS	Fuel countdown expires AND rescue shuttles > 0 (auto-dispatch)

INVALID TRANSITIONS

The following transitions must NEVER occur:

- GAME_OVER → any state (terminal state — requires new game/restart)
- LAUNCH_SEQUENCE → REFIT_AND_REWARD (cannot skip combat)
- TEST_FLIGHT → ACTIVE_COMBAT (test flight cannot transition to a real mission)
- DOCKING_CUTSCENE → ACTIVE_COMBAT (cannot return to combat after docking)
- RESCUE_IN_PROGRESS → GAME_OVER (rescue always completes if a token was available)

8. Round Difficulty Scaling

Complete round-by-round progression details, enemy specifications, ability unlock schedules, and rank thresholds are defined in **BFXG-008 (Progression Bible)**. This section summarizes how the mission cycle itself adapts across the 20-round arc.

Round Range	Tier Name	Mission Cycle Adaptations
Rounds 1–2	Tutorial / Cadet	No boss encounters Simplified briefings with tutorial language explaining HUD and controls Fewer enemy waves, slower enemy AI Extended fuel supply (reduced depletion rate) Mandatory first docking cut scene (unskippable) Briefing explicitly

Round Range	Tier Name	Mission Cycle Adaptations
		introduces the rescue shuttle system
Rounds 3–5	Ensign	First boss gate at Round 3 Fuller briefings — threat descriptions become more detailed Secondary objectives introduced Wingman chatter becomes more frequent Standard fuel depletion rate
Rounds 6–9	Lieutenant	Multi-wave missions — 2–3 enemy waves per round Boss encounters every other round More complex objectives (escort + destroy combos) Briefings include tactical recommendations Enemy formations become more coordinated
Rounds 10–14	Commander	Multi-phase missions — objective changes mid-round Faction shifts — new enemy types with different AI behaviors Elite briefings — announcer tone becomes more urgent Fuel management becomes critical (faster depletion) Enemies deploy countermeasures against the player
Rounds 15–19	Captain / Admiral	Stealth missions — radar-silent approaches, reduced HUD Capital ship battles — player must disable subsystems in sequence Minimal HUD variants — some elements disabled for immersion Briefings are terse, high-stakes — "You know the drill, pilot."

Round Range	Tier Name	Mission Cycle Adaptations
		Every round has a boss or mini-boss
Round 20	Fleet Admiral — FINAL	Extended briefing — narrative weight, references to the full journey All player abilities and weapons available Multi-phase final boss with unique mechanics All emergency systems active but under extreme pressure Victory triggers the ending cut scene and credits roll

9. Integration with ButterflyFX ANPC Design Suite

This document (BFXG-009) is the ninth entry in the ButterflyFX ANPC Design Suite. The table below maps how this document connects to every other document in the suite.

Doc ID	Document Title	Integration with BFXG-009
BFXG-001	General ANPC Guideline	Defines NPC behavior during dock scenes — crew members in the docking bay (Phase 6), repair drones during refit (Phase 7), and ambient NPC activity in the launch bay environment.
BFXG-002	Starfighter ANPC Guideline	Governs Wingman AI behavior during active combat (Phase 3) — tactical callouts, formation flying, combat engagement rules. Also governs enemy AI

Doc ID	Document Title	Integration with BFXG-009
		formations, attack patterns, and difficulty scaling behaviors.
BFXG-003	Fast Track ANPC Guideline	*N/A — Different game (Fast Track racing title). No direct integration.*
BFXG-004	Schwartz Diamond	*N/A — Different game (Schwartz Diamond title). No direct integration.*
BFXG-005	Voice Guide	Defines the Launch Bay Announcer voice profile (authoritative, military cadence), Wingman voice character (energetic, personal), and Command voice character (calm, professional). All voice lines in Section 3 must conform to BFXG-005 voice direction.
BFXG-006	GLB Optimization	Governs model loading and LOD strategies for the HumanSpaceBattleShip.glb (carrier — large, complex model requiring LOD management), Rescue Shuttle.glb, FuelTanker.glb, and all environment assets. Critical for docking cut scene performance (Phase 6) where the full carrier interior is visible.
BFXG-007	Manifold Multipliers	Runtime tessellation system used for the carrier exterior (dynamic detail as player approaches during Phase 5/6), space environment geometry (asteroids, debris), and launch tube interior surfaces. Ensures the "Store Geometry Not Bytes" philosophy is maintained.

Doc ID	Document Title	Integration with BFXG-009
BFXG-008	Progression Bible	The primary companion document to BFXG-009. Defines: round-by-round enemy compositions, boss specifications and health pools, ability/weapon unlock schedule (what is awarded at each round), rank thresholds and titles, XP/credit formulas, and difficulty curve parameters. The mission cycle (this document) consumes data defined in the Progression Bible at every phase.

DEPENDENCY HIERARCHY

BFXG-008 (Progression Bible) provides the DATA. **BFXG-009** (this document) provides the LOOP. Together they define the complete player experience. Neither document is sufficient alone — developers must reference both when implementing any round.

10. Appendix: Complete Mission Cycle Implementation Checklist

Use this checklist to track implementation progress. Every item must be completed for the mission cycle to function. Items are organized by phase.

10.1 Phase 1 — Launch Bay Briefing

☐	Implementation Item
☐	Launch Bay interior environment loaded and rendered (LaunchBay_Interior.glb)
☐	Cockpit interior camera positioned in launch tube (pilot POV)
☐	Announcer briefing system — parameterized script engine with variation selection
☐	Threat briefing data feed from round configuration (BFXG-008)
☐	Primary objective display on HUD
☐	Secondary objectives display
☐	Weapon/ability loadout display (current inventory)
☐	New unlock announcement trigger (conditional)
☐	Rank achievement announcement trigger (conditional)
☐	Clearance-to-launch line — triggers transition to Phase 2
☐	Round 1 tutorial variant of briefing

10.2 Phase 2 — Launch Sequence

☐	Implementation Item
☐	Launch tube door open animation
☐	Klaxon alarm and amber warning lights
☐	Catapult rail electromagnetic effect (blue energy on tube floor)
☐	Fighter acceleration down tube with G-force visual effects
☐	Camera transition: cockpit POV → external chase cam
☐	Carrier receding in background

☐	Implementation Item
☐	HUD power-on sweep animation (sequential element activation)
☐	Player control handoff after HUD activation
☐	Skip functionality (hold SKIP 1 second, disabled on Round 1)
☐	Audio: klaxon, catapult charge, acceleration roar, space transition, engine hum

10.3 Phase 3 — Active Combat

☐	Implementation Item
☐	Full flight control system (thrust, pitch, yaw, roll, afterburner)
☐	All 18 HUD elements implemented and positioned (see Section 4)
☐	Enemy spawning system tied to round configuration
☐	Primary objective tracking with completion detection
☐	Secondary objective tracking
☐	Command voice — enemy wave alerts (parameterized)
☐	Command voice — objective progress updates
☐	Command voice — boss appearance warning (Round 3+)
☐	Wingman voice — tactical callouts, encouragement, warnings
☐	Boss health bar activation (when boss engaged)
☐	Shield regeneration system
☐	Fuel depletion system (baseline, afterburner 3x, idle 0.25x)

☐	Implementation Item
☐	Weapon firing, ammo tracking, cooldown timers
☐	Damage model: shield absorption → hull damage
☐	Score/credit accumulation per kill and objective
☐	XP accumulation system

10.4 Phase 4 — Emergency Systems

☐	Implementation Item
☐	**Rescue Shuttle:** Hull = 0 detection → rescue trigger (if tokens > 0)
☐	Rescue Shuttle model loading and approach animation
☐	10-second rescue cut scene (tractor beam, docking, repair, reboot)
☐	Full ship restoration (hull, shields, ammo, fuel)
☐	Safe re-entry repositioning logic
☐	Rescue shuttle token decrement and HUD update
☐	Last-shuttle warning announcement
☐	**Fuel System:** Fuel gauge with 3-tier color coding (green/yellow/red)
☐	Low fuel alarm at 25% threshold
☐	RED ROUND FUEL BUTTON — UI element, pulsing when fuel < 25%
☐	Fuel tanker dispatch, approach, docking, and 5-second hold timer
☐	Sharp-turn detection to break/reset refueling connection
☐	Refueling call limit: 2 per round, with remaining-calls indicator

☐	Implementation Item
☐	Dead stick state at 0% fuel (drift, no thrust, weapons active)
☐	30-second countdown at 0% fuel with no refuel calls → auto-rescue or Game Over
☐	**Countermeasures:** Missile lock detection and alarm
☐	Incoming missile warning HUD overlay with directional indicator
☐	Flare/chaff deployment with limited supply tracking
☐	Countermeasure success/fail resolution logic

10.5 Phase 5 — Mission Complete

☐	Implementation Item
☐	Primary objective completion detection → mission complete trigger
☐	Mission complete announcement (voice line selection)
☐	Victory fanfare audio sting
☐	Player control disable → autopilot state
☐	AUTOPILOT indicator on HUD
☐	Ship auto-navigation toward carrier
☐	Camera-look rotation (player can look around, no flight control)
☐	Environmental scenery during return (debris, allies, celestial bodies)
☐	15–20 second return flight duration

10.6 Phase 6 — Docking Cut Scene

☐	Implementation Item
☐	9-step docking sequence with camera transitions (see Phase 6 table)
☐	Carrier exterior model with landing lights
☐	Docking bay door animation
☐	Docking bay interior (parked fighters, crew, equipment)
☐	Landing gear deployment animation
☐	Docking clamp engagement animation
☐	Rail system movement to launch tube
☐	All docking SFX (hydraulics, clamps, gear, rails, engine wind-down)
☐	Skip functionality (hold 1 second, disabled Round 1)

10.7 Phase 7 — Refit & Reward

☐	Implementation Item
☐	Welcome back announcement
☐	Animated refit sequence (fuel lines, repair drones, ammo loading, countermeasure restock)
☐	Performance summary display (all 9 metrics from Section 1.8)
☐	XP calculation and rank progress bar update
☐	Credit calculation and running total display
☐	New ability/weapon unlock announcement (conditional)
☐	Ability/weapon icon and description

☐	Implementation Item
	display
☐	Loadout auto-update with new item
☐	Rank promotion ceremony animation (conditional)
☐	Rank insignia display and fanfare
☐	Refit status announcement
☐	Ship fully restored: hull 100%, shields 100%, fuel 100%, ammo full, countermeasures restocked

10.8 Phase 8 — Test Flight

☐	Implementation Item
☐	"TEST FLIGHT" button UI element in refit screen
☐	Test flight invitation announcement
☐	Abbreviated launch sequence (5 seconds)
☐	Test environment: open space near carrier
☐	Stationary target dummies with damage number display
☐	Moving target drones with AI flight patterns
☐	Obstacle course with marker gates
☐	Unlimited ammo and fuel in test mode
☐	"TEST MODE" watermark on HUD
☐	"RETURN TO LAUNCH BAY" button
☐	"SKIP TO MISSION" button
☐	Abbreviated return/docking sequence (8 seconds)
☐	No time limit enforcement

10.9 Phase 9 & 10 — Next Briefing & Relaunch

☐	Implementation Item
☐	Round counter increment
☐	Next round data load from progression configuration (BFXG-008)
☐	Announcer briefing with updated parameters (threat, objectives, loadout)
☐	Launch sequence replay
☐	State machine cycle reset to ACTIVE_COMBAT after launch
☐	Per-round resource reset: refueling calls = 2, countermeasures restocked
☐	Persistent resources carry over: rescue shuttle tokens (NOT reset per round)

10.10 Game Over & Victory

☐	Implementation Item
☐	Game Over detection: all 3 conditions (GO-1, GO-2, GO-3)
☐	Ship destruction cut scene (canopy cracks, explosion, debris, fade to black)
☐	Ejection cut scene (canopy jettison, seat rocket, beacon, fade to white)
☐	Game Over voice lines
☐	Final score screen (rank, rounds completed, enemies destroyed, accuracy, credits)
☐	"PLAY AGAIN" and "MAIN MENU" buttons
☐	Ejection mercy mechanic: rank and credits saved to persistent profile

☐	Implementation Item
☐	Victory condition: Round 20 completion → ending cut scene → credits roll
☐	Pause menu with Eject option (confirmation dialog required)

10.11 Cross-Cutting Systems

☐	Implementation Item
☐	State machine with all 11 states and valid transitions implemented
☐	Invalid transition prevention (see Section 7.2 warning)
☐	Announcer script engine: parameterized templates, random variation selection, no-repeat logic
☐	Voice line audio playback system (3+ variations per category, 17 categories)
☐	Music system: combat intensity, victory calm, refit ambient, ceremony fanfare
☐	SFX library: 20 sound effect categories loaded and mapped
☐	GLB model loading per BFXG-006 optimization guidelines
☐	Manifold Multiplier tessellation for carrier and environment (BFXG-007)
☐	Cut scene skip system (hold 1 second, Round 1 restrictions enforced)
☐	Difficulty scaling tied to round number (see Section 8)
☐	Persistent game data: rescue shuttle tokens, rank, credits, unlocked abilities
☐	Per-round reset data: refueling calls, countermeasures, fuel, ammo

BFXG-009 — Starfighter Launch Bay & Mission Cycle Design Document

ButterflyFX ANPC Design Suite — Version 1.0

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"Store Geometry Not Bytes" — ButterflyFX